Algorithms

Subject: Computer Science | NoteShare Library

ALGORITHMS

1. COMPLEXITY
O(1) < O(log n) < O(n) < O(n log n) < O(n) < O(2^n)

2. MASTER THEOREM
T(n) = aT(n/b) + f(n)
3 cases based on n^(log_b a)

3. DIVIDE & CONQUER
Merge Sort O(nlogn)
Quick Sort O(nlogn)
Binary Search O(logn)
Strassen O(n^2.81)

4. GREEDY
Activity Selection
Huffman Coding
Dijkstra, Prim, Kruskal
Fractional Knapsack

5. DYNAMIC PROGRAMMING
Fibonacci O(n)
0/1 Knapsack O(nW)
LCS O(mn)
Edit Distance O(mn)
Matrix Chain O(n)
Coin Change

6. GRAPH
BFS O(V+E)
DFS O(V+E)
Dijkstra O((V+E)logV)
Bellman-Ford O(VE)
Floyd-Warshall O(V)
Topological O(V+E)

7. NP-COMPLETENESS
P, NP, NP-Complete, NP-Hard
SAT, 3-SAT, Clique, TSP

8. APPROXIMATION
TSP: 2-approx
Vertex Cover: 2-approx
Set Cover: O(log n)